

PC

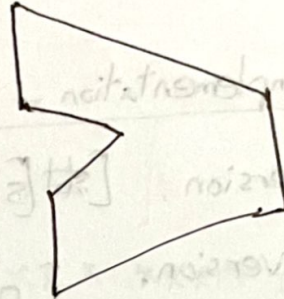
#8000

8001	XXXX XXx0
8002	0101 1010
8003	0XXX 0011
8004	0XXX XXXX
...	
8009	

instruction
memory

XXXX XXx0 0101 1010 0XXX 0011
 func7 rs2 rs1 func3 rd
 opcode

(V-0279)



Topics:-

i) Instruction fetch

(ii) " Decode

(iii) Reg file read

(iv) ALU

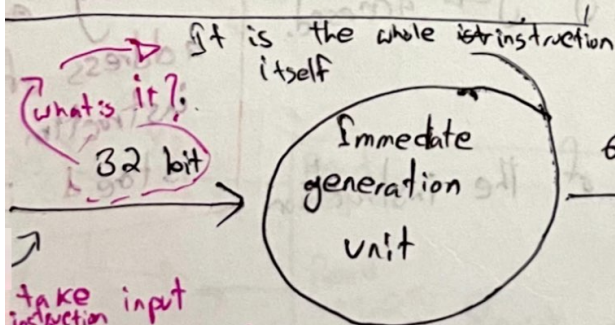
(v) Data Memory access

(vi) write Back

X₁₂ = 0x1234AB789ABCDEF01

Actual A

ALU

take input
instruction
not the hex val.then it will extract
the 12 bit. Then it will expand 12 bit to 64 bit.

Actual address = Base + Offset

1010 1100 1110 1111

1110 1110 1100 1100

To know about the CPU time we have to know the Instruction Count (IC). To find out the IC we have to know about the ISA on the language we are using for the program (RISC-V).

RISC-V implementation - 2 types:-

- (i) Simplified version. [slow]
- (ii) Pipelined version. [optimized]

INSTRUCTION EXECUTION STEP:-

Instruction Memory :- Holds the instructions only

Data Memory : Main memory. In case of load/store this memory gets accessed.

STEPS:-

(i) PC contains the address of the instruction in the program being executed.

Using this address from PC we fetch the instruction from the instruction memory.

PC:- Program Counter register

The instruction which we are currently executing and the address of that instruction are stored in PC

(ii) The fetched instruction gets decoded. Then the registers which associated with the decoded instruction will become activated so that it can read data/write data.

Example :-

PC = #6000

Instruction Memory

#6000	xxxx xxx 0
#6001	0101 1010
#6002	0xxx 0011
#6003	0xxx xxx x

step-1

step-2

XXXX XXXX 0101
0XXX 0011 0XXX XXXX

Decodes step-3

By & By decoding we try figure out the source of the registers

Every instruction size is 32 bit and every cell of the IM is 8 bit.

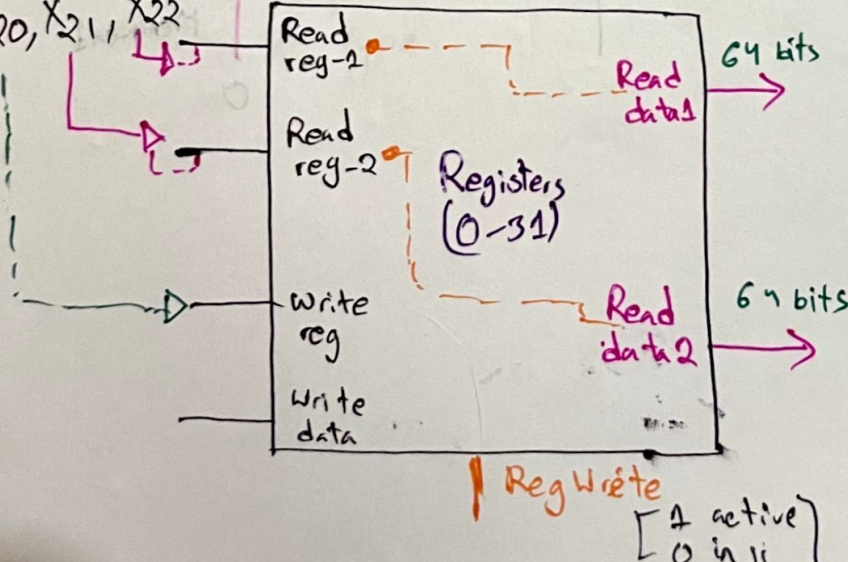
So 4 cell required to store 1 instruction

That's why: PC+4

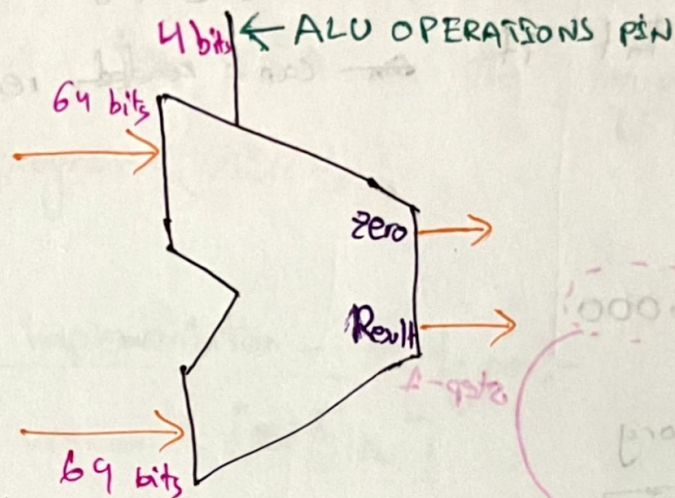
#REGISTER FILE

In a register file the 32 registers are stored.

add X20, X21, X22



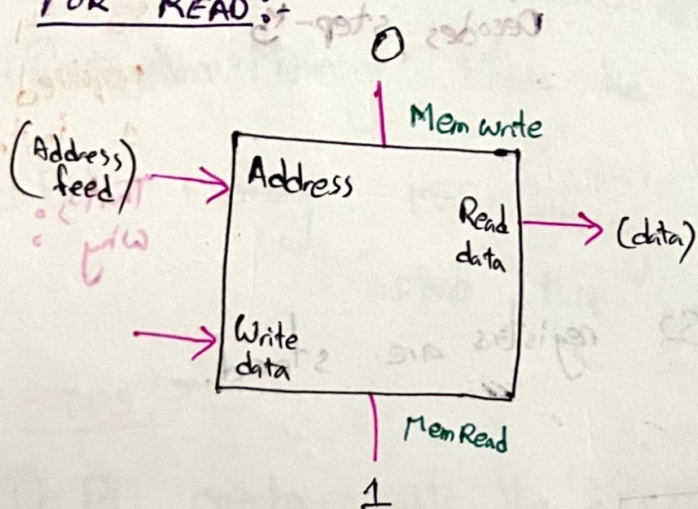
Arithmetic Logic Unit (ALU)



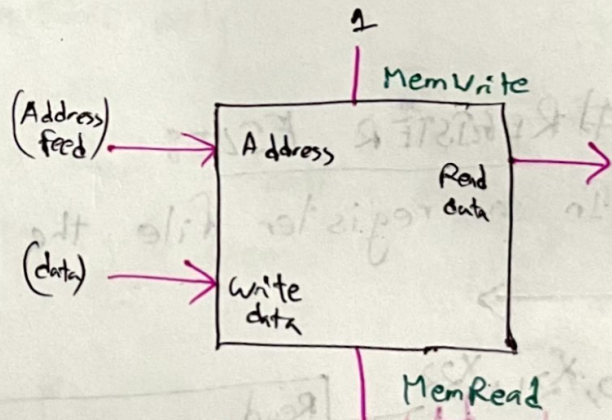
ALU tells that if the result is 0 or not

DATA MEMORY UNIT :-

For READ :-



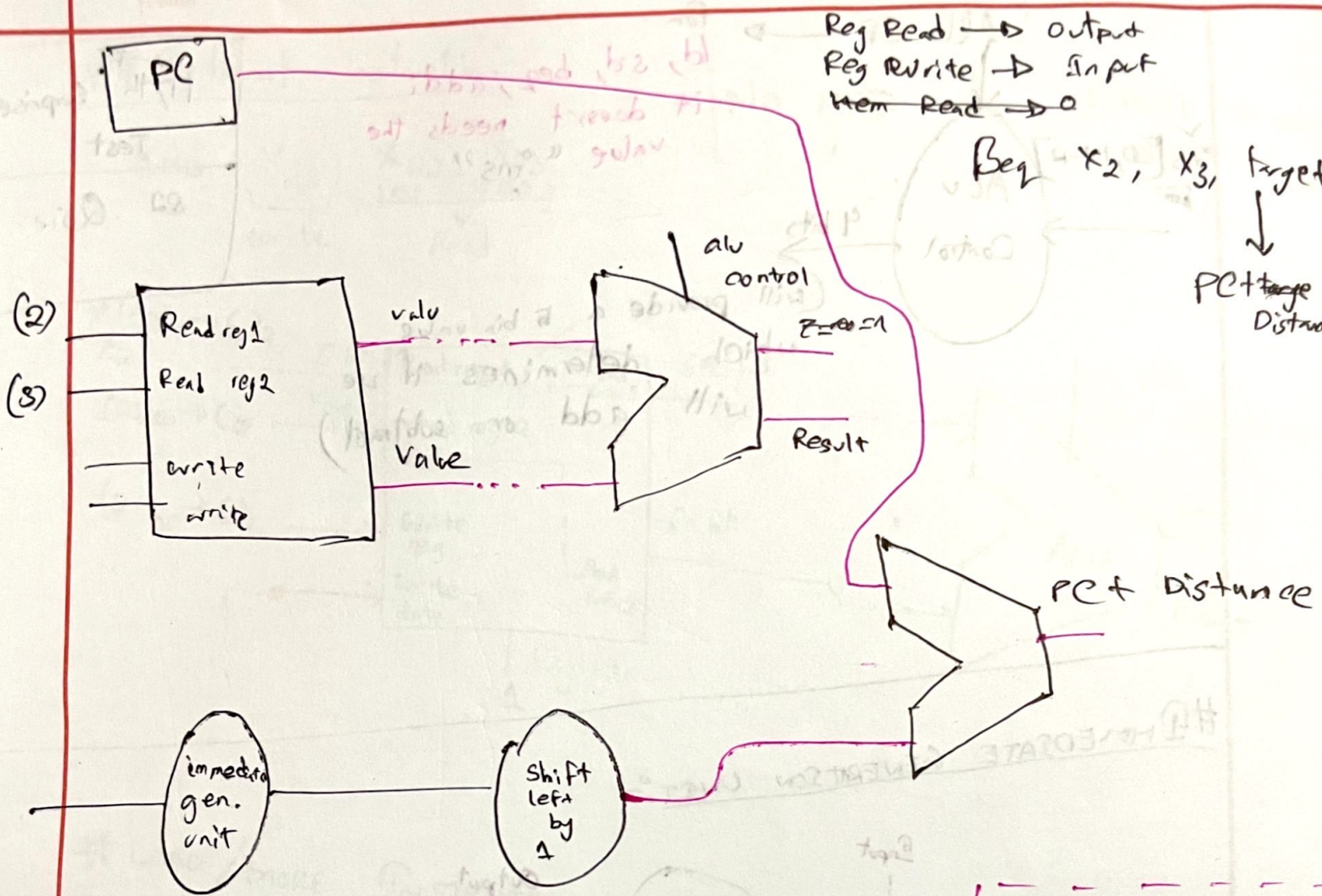
For WRITE :-



Subject :

Mem Write $\rightarrow$ Input } Memory
Reg Read $\rightarrow$ output
Reg Write $\rightarrow$ Input
Mem Read $\rightarrow$ 0

Date: 10/09/24
Time:



ALU OP $\Rightarrow$

- 00 - load / store / addi
- 01 - Beq
- 10 - R type

Q. type

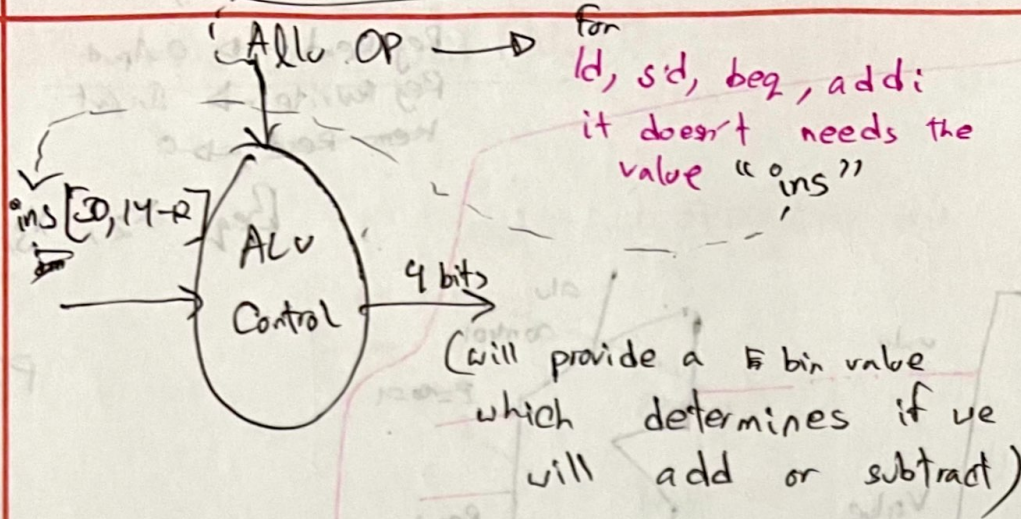
what is the value generated in different pins of control unit?

Subject :

Control unit

Date :

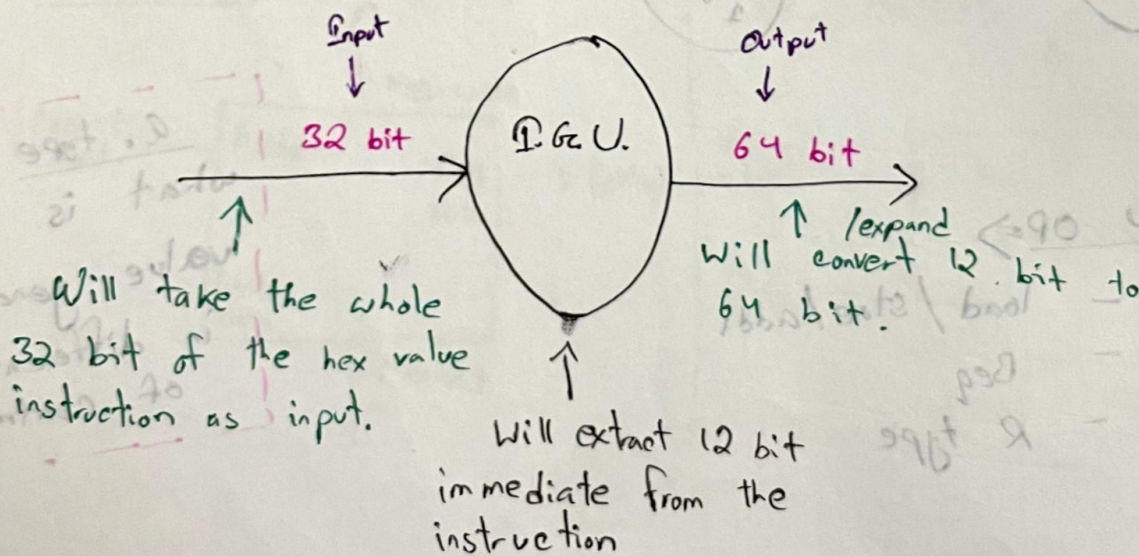
Time :



17th Surprise Test

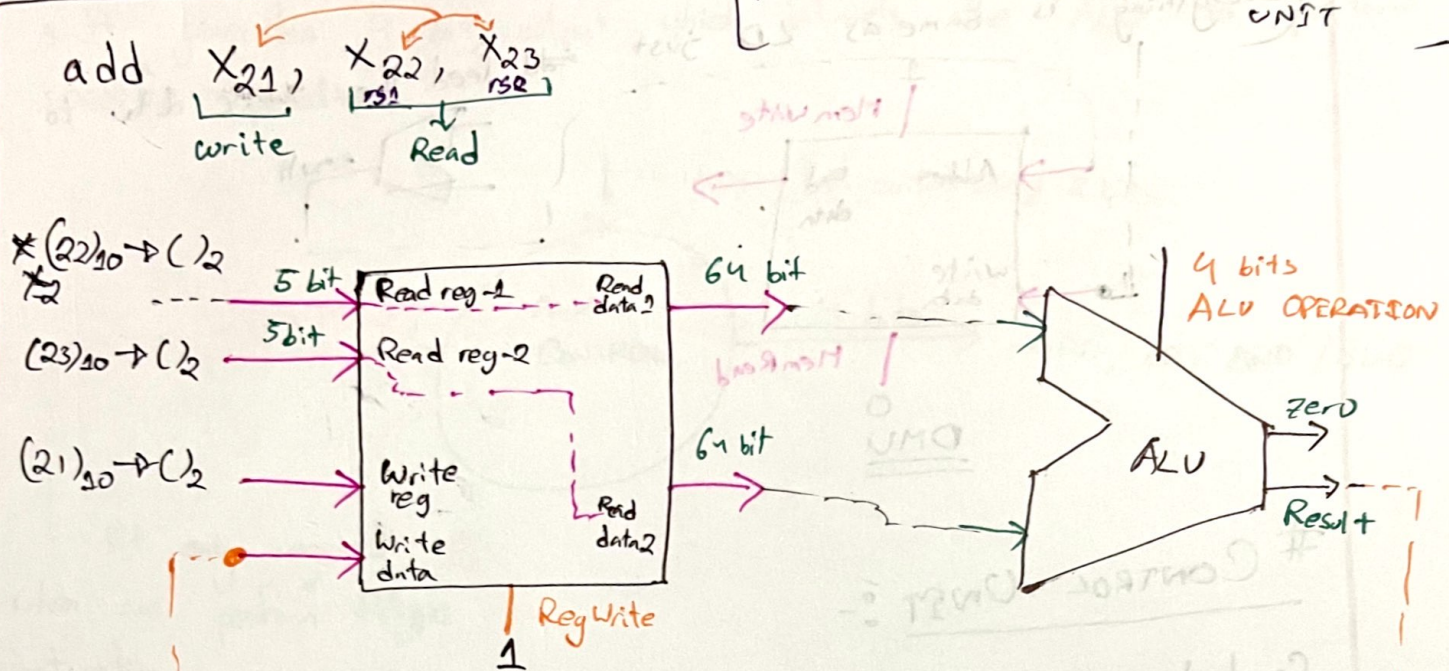
22 Quiz

IMMEDIATE GENERATION UNIT :-

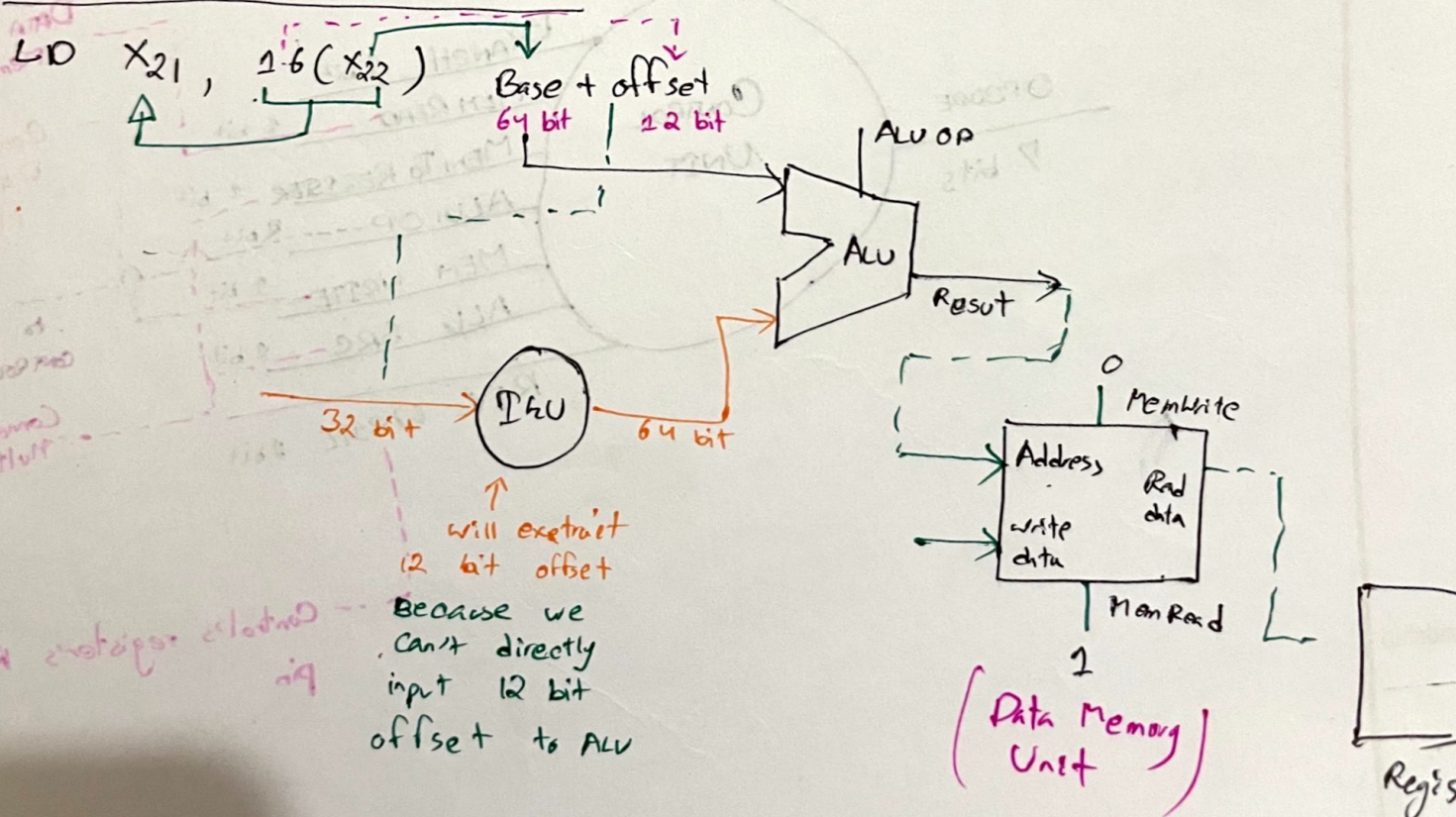


R-format instruction:-

[NO NEED OF DATA MEMORY UNIT]

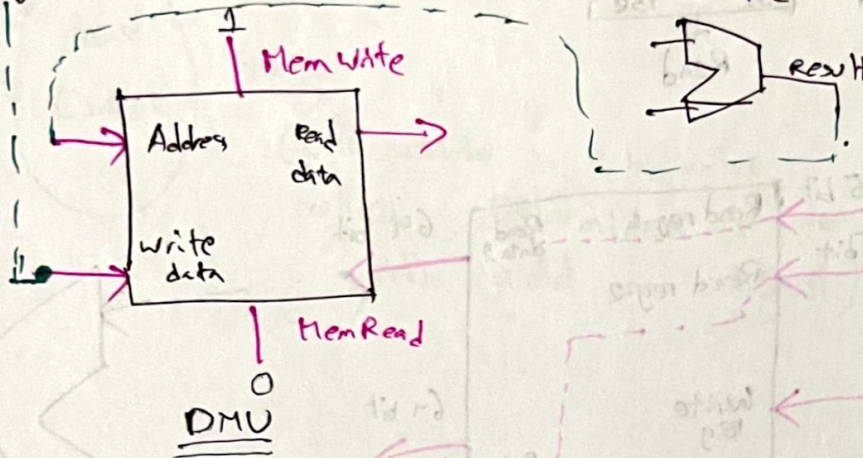


LOAD / STORE INSTRUCTION:



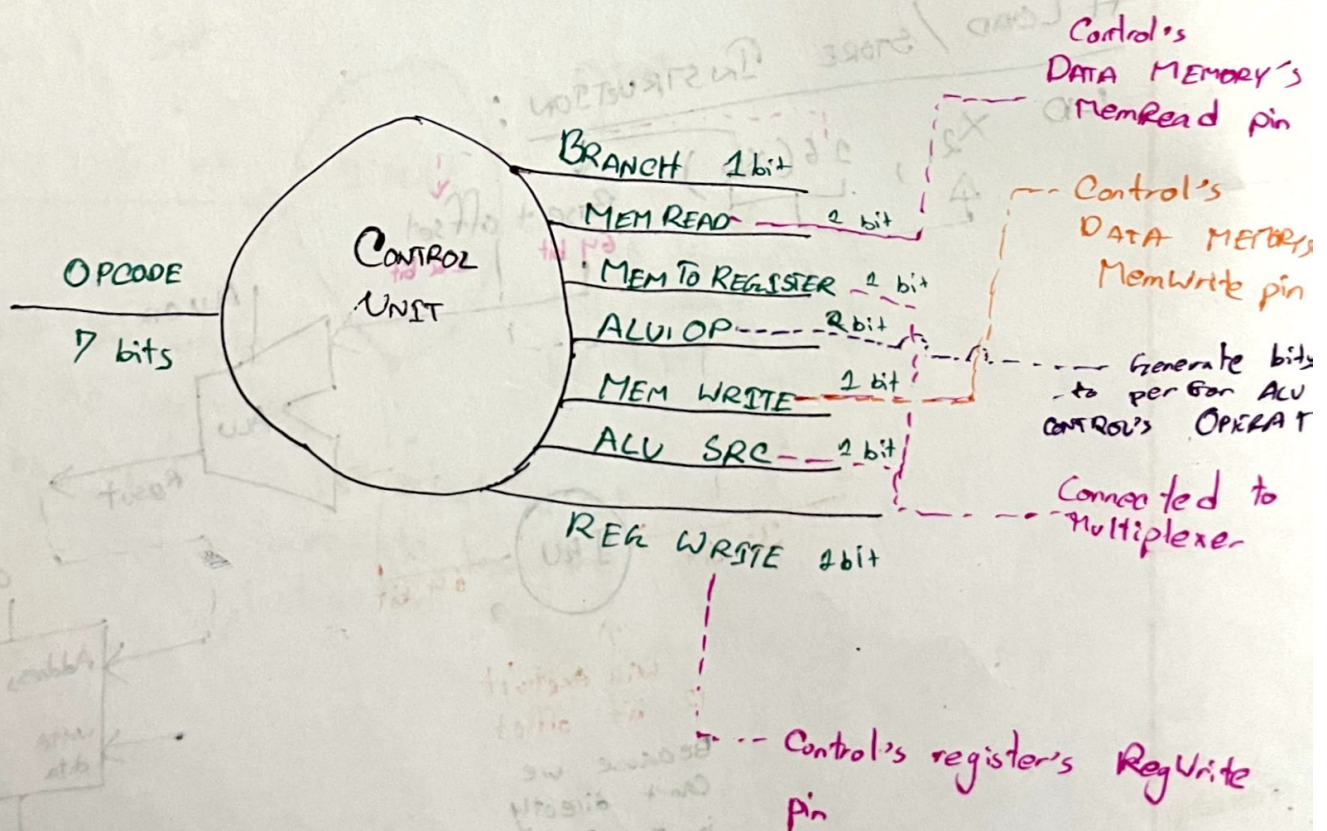
SD $X_{21}, 16(X_{23})$.

(everything is same as LD just add feed the write data to DMU)



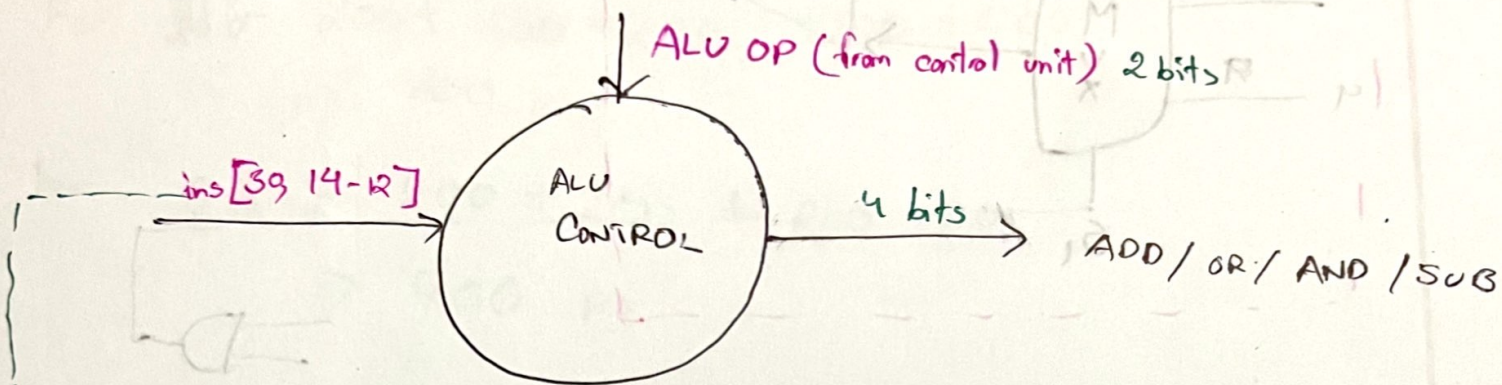
CONTROL UNIT :-

Control unit handles the working procedure of ALU control:



ALU CONTROL

It generates 4 bit output which instructs the ALU to perform specific operations.

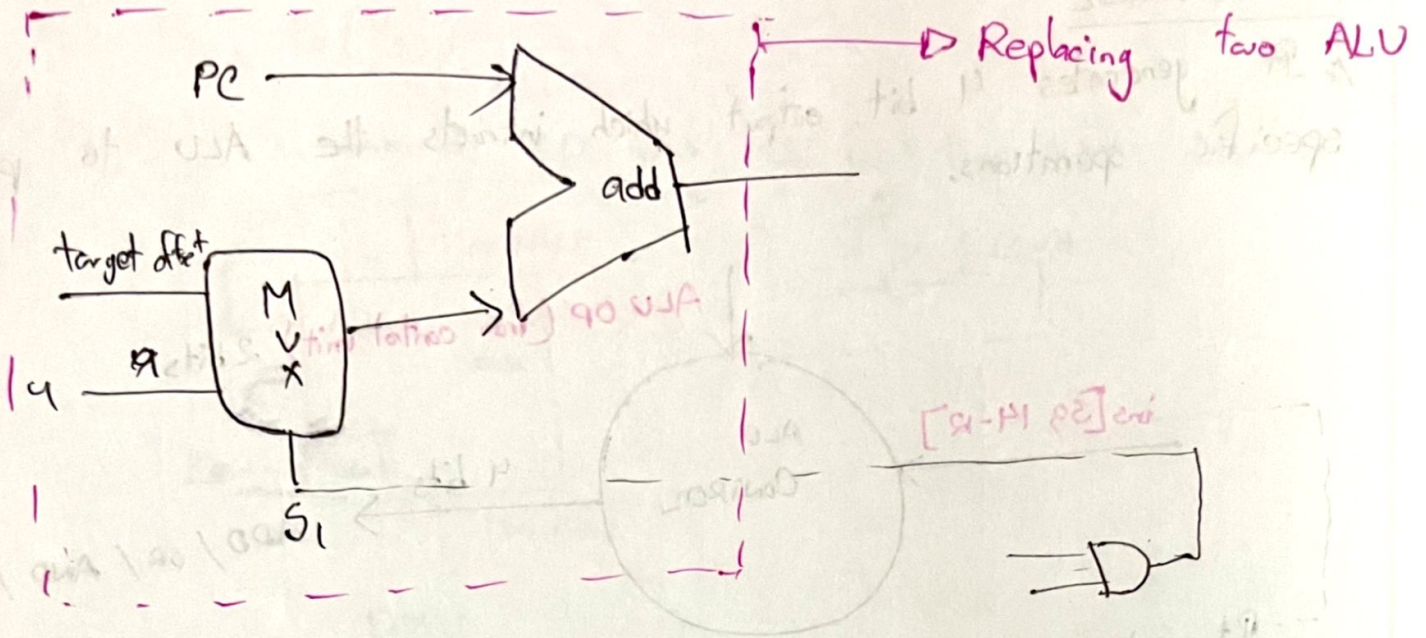


It only requires when we perform R-type instruction.

only { X₂, X₃, X₁, X₂, X₃, X₁ }

66A

66A



★ Mux instruction input reversed $\rightarrow$ R type & I type. Then what will be the signal of ALU src.

★ Draw the data path for

Add X_2, X_3, X_4

Addi X_3, X_3, X_4

only

$\rightarrow$ draw the selected pins.

★ Might be question which where we will replace a component from the data path. At The q. will ask for if the data path or not.

PIPELINE :-

Performance :-

For ~~Id~~ :- $200 + 100 + 200 + 200 + 100$
 $\Rightarrow 800 \text{ ps}$

For ~~Sd~~ :- $200 + 100 + 200 + 200$
 $\Rightarrow 700 \text{ ps}$

For R-format :- $200 + 100 + 200 + 100$
 $\Rightarrow 600 \text{ ps}$

For ~~Beq~~ :- $200 + 100 + 200$
 $\Rightarrow 500 \text{ ps}$

For single ^{cycle} ~~cycle~~ what will be the clock period?

$\rightarrow 800 \text{ ps}$

[Longest delay - determines the clock period]

clock period = 800 ps

Total time = $800 \times 4 = 3200 \text{ ps}$

for all instruction

Total time loss = $800 + 700 + 600 + 500$
 $= \cancel{3200}$
 $= 600 \text{ ps}$



PIPELINE :-

2# Different executions of instruction ^{are} gets overlapped. [Parallelism occurs]

Different instructions are executed at the same time.

At a single clock cycle, we run/execute 1 instruction

NOTE :-
Execution of instructions into several stages, allowing multiple instructions to proceed simultaneously

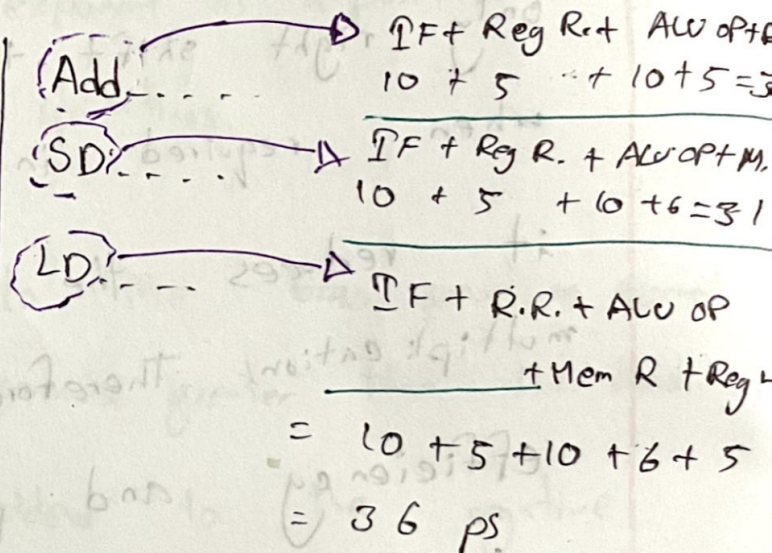
EXAMPLE :-

Instruction fetch = 10 ps

Register file read/write = 5 ps

Memory read/write = 6 ps

ALU Operation = 10 ps



CLOCK PERIOD OF PIPELINED $\rightarrow$ largest delay of a stage

Highest time for instruction = 36 ps

CLOCK PERIOD OF SYSTEM :- longest delay to run an instruction

clock period = 36 ps

Clock Cycles

Total time to run a instruction sequence = clock period $\times$ instruction

single cycle means = 1 instruction per clock cycle.

N_{or} of clock cycles = stage + (n-1) [Piplined]

n = no. of instruction

STRUCTURAL HAZARD:-

If at the same clock cycle we are using a single instruction step ^{/Process} many times more than 1 times

NOTE:-

At a single clock cycle we cannot use a single clock cycle more than 1 time

★ Why we have separate memory for instruction storing & data storing?

→ In order to avoid structural hazard we have divided our memory into two ^{different} categories.

- Instruction memory
- Data "

DATA HAZARD:-

When try to decode or read data from a certain register before its data gets updated then its ~~data~~ completion of update data is hazard. (Because, then we are just reading a garbage value from that register.)

Ways to solve data hazard:-

- (i) STALLING
- (ii) FORWARDING
- (iii) ^{CODE} SCHEDULING

There is no data hazard in here

SD X22, 10[X21]
Add X22, X22, X21

Because, the ~~data~~ data isn't getting updated

Subject :

Date :

Time :

